



An inclusive school for computer science: Evaluating early impact with propensity score matching

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ABSTRACT

Inclusive STEM high schools are designed to address racial, gender, and socioeconomic disparities in access to high-quality STEM education. Our study focuses on an inclusive, computer-science-themed, regional high school in Virginia, which admits students via a weighted random lottery. Using a quasi-experimental design, we analyzed a restricted-use, longitudinal dataset to compare outcomes for admitted students to those of a propensity-matched group of non-admitted applicants who attended traditional schools. Results from the school's first two years show positive effects of admission on computer science course-taking and attendance, but negative effects on several standardized test outcomes. Our study contributes to research examining the potential of schools designed for equity and centered on STEM, and underscores the importance of careful analytic approaches to evaluation.

Recent decades have seen an increase in awareness that women and members of minoritized racial and ethnic groups are severely underrepresented in STEM (science, technology, engineering, and math) fields, in comparison to white and Asian men. For example, Black employees currently make up nearly 12% of the total U.S. workforce, but hold only 8.5% of STEM jobs and just 7% of STEM jobs requiring a bachelor's degree. For Latinx people, who are 18% of the workforce, the figures are 14%, and 8%, respectively (National Science Board, 2021a). Women are overrepresented in health-related STEM jobs (e.g., dental hygienists, nurses, medical records and health information technicians), but vastly underrepresented in many other STEM occupations (e.g., in computing fields, mathematics, engineering, and physical science; Pew Research Center, 2021). Similar patterns of underrepresentation prevail in STEM degree-earning, at every educational level (Pew Research Center, 2021), and are evident in high school course- and exam-taking as well (National Science Board, 2021b; Office for Civil Rights, 2018).

Policymakers, educators, and others have expressed concern about these patterns from an equity perspective (e.g., NASEM, 2021a; Virginia Governor's STEM Education Commission, 2020); from the perspective of likely implications for the U.S.'s global competitiveness in technology,

as well as its national security and economic prosperity (e.g., NASEM, 2007; NASEM, 2011); and from a perspective that understands diversity in STEM as a crucial asset for innovation and leadership (e.g., Harris, 2021; NASEM, 2011). Such concerns have prompted efforts to improve historically underrepresented students' access to high quality secondary STEM education, and to increase the appeal of STEM majors and careers to those students.

One type of initiative has been the establishment of "inclusive STEM schools": STEM-focused schools intended to help address racial/ethnic and socioeconomic disparities in access to STEM education opportunities (Gamoran, 2016; Means et al., 2017; National Research Council, 2011; PCAST, 2010). Because inclusive STEM high schools are a recent development, systematic data on the schools' outcomes are fairly limited. A series of studies in North Carolina, Texas, and Ohio by Means and colleagues provides some evidence of the schools' potential for promoting students' STEM career interest and aspirations, advanced course-taking, and further participation in STEM (Means et al., 2016, 2017, 2018, 2021; see also Bicer et al., 2020).

Scholars have found only limited positive effects of inclusive STEM schools on student achievement, however (Gnagey & Lavertu, 2016;

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Means et al., 2017; Means et al., 2021; Saw, 2019). Another line of investigation sounds a stronger cautionary note, highlighting the challenges of sustaining programs at inclusive STEM schools when they compete with other educational priorities. For example, Weis et al.'s (2015) longitudinal, ethnographic study of four STEM-focused schools identified "mechanisms of erosion" (p. 1050) that undermined the schools' vision of making high-level STEM available to all students. They found that a combination of district graduation requirements, accountability pressures such as standardized test pass rates, and related class size and staffing issues hampered expansion of STEM course offerings, ultimately leading the schools to reduce the advanced STEM courses available and even to limit advanced content coverage in courses they did offer.

Our case study centers on CodeRVA Regional High School – a new school opened in 2017 and designed to help address inequities in computer science education and minority group isolation, both of which have resulted from persistent school segregation in central Virginia. Residential segregation and a lack of regional cooperation have hobbled school desegregation efforts in central Virginia since the 1970's, creating a situation where not only individual schools but entire school systems are largely segregated (Siegel-Hawley, 2020). CodeRVA attracts students across participating school district boundaries, bringing together a diverse student body intended to be representative of the region, and providing a full computer science education for students who might not otherwise have access to it.

Most studies of the effectiveness of inclusive STEM schools have relied on selection by the researchers of inclusive STEM and comparison schools in the same state or region, a strategy that introduces potential unmeasured confounding variables due to the organization or operation of the schools themselves. Equally important, it also limits the possibility of fully accounting for potential differences between students in the treatment and comparison groups, since students generally choose to attend inclusive STEM schools and may differ from students at traditional comprehensive schools in ways that are unmeasured by the researchers. For example, students who choose to attend STEM-focused schools may be more motivated, or feel more interested in or connected to STEM than students who attend traditional schools (as acknowledged, for example, in studies by Means et al., 2017, 2021). In our case, however, all students are part of a single applicant pool, and treatment assignment occurs at the student level. We take advantage of CodeRVA's oversubscribed, weighted random admissions lottery to carry out a quasi-experimental comparison of outcomes for applicants assigned to admission and those not assigned, who instead attended mostly traditional comprehensive schools in their home school districts. Further, we use propensity score matching to adjust for differences (due to weights affecting selection probabilities) between admitted and non-admitted applicants. Finally, we leverage Virginia's statewide longitudinal data system to obtain school systems data for all CodeRVA applicants beginning in middle school, allowing us to take account of prior academic outcomes.

Our investigation centers on three different dimensions of CodeRVA's potential impact during 2017–2019, its first two academic years. First, we examine the school's effect on computer science course-taking. Second – bearing in mind that one goal of specialized STEM schools is to get students interested in and excited about STEM – we examine the school's impact on attendance, treating it as a behavioral measure of engagement. Third, we examine CodeRVA's effect on academic outcomes in the form of scores on the standardized tests students take at the end of math, science, and social science courses during their first several years of high school. Guided by the school's goal of reducing opportunity gaps,¹ we also consider differential effects by race, socioeconomic status, gender, and residence in an urban or non-urban school district for each aspect of the school's potential impact.²

1. Inclusive STEM schools: definition and background

Inclusive STEM high schools are designed "to develop students' STEM expertise rather than to select those students with prior demonstration of STEM talent" (Means et al., 2016, p. 710). Some schools focus on a particular STEM area, and others emphasize STEM more broadly; more generally, they do not adhere to a single model or approach. They do, however, share the goal of expanding the population of students with STEM-related skills by increasing access to specialized education for historically underrepresented students, especially female students, Black and Latinx students, and/or those from economically disadvantaged backgrounds. To advance toward these goals, inclusive STEM schools enroll students based on their interest, rather than screening students for aptitude or prior achievement; provide more intensive STEM preparation than traditional comprehensive schools; and provide supports to prepare all students (not just a subgroup) to pursue further STEM work or education (Bicer et al., 2020; LaForce et al., 2016; Lynch et al., 2018; Means et al., 2021; Peters-Burton et al., 2014; Waters & Orange, 2022).

School initiatives that ensure equitable access and preparation are particularly important in light of evidence that disparities in STEM participation, performance, and career aspirations begin in students' early formal education experiences (Betancur et al., 2018; Henry et al., 2020; Morgan et al., 2016, 2022; Quinn & Cooc, 2015; Riegle-Crumb et al., 2011). Research on STEM education has identified multiple sources of these disparities. Reviews by Xie et al. (2015) and Bottia et al. (2021) discuss structural barriers that limit K-12 STEM participation, preparation, and engagement by students from lower socioeconomic status backgrounds and – because class and race/ethnicity are closely correlated – by racially or ethnically minoritized students. Such barriers include, for example, under-resourced schools and lack of access to high-quality, challenging math and science classes, as well as disproportionate tracking of minoritized students in academic pathways that restrict their participation in advanced math and science courses.

In addition, educational contexts can be unwelcoming for historically underrepresented students, with consequences for their engagement and persistence in STEM. For example, curriculum that privileges Western, white ways of knowing, environmental cues that signal to racially minoritized students and/or girls that they do not belong, and implicit biases or microaggressions can reduce students' engagement and sense of belonging in STEM (e.g., Bang et al., 2012; Bottia et al., 2021; Master et al., 2016; Tzou et al., 2021). Conversely, environments that include plentiful opportunities to engage with high-level curriculum, employ culturally relevant instruction, and/or that value students' home and community knowledge bases can enhance STEM engagement, belonging, and intended persistence (e.g., Bae, Sealy, Cabrera, Gladstone, & Mills, 2022; Bicer et al., 2020; Calabrese Barton & Tan, 2009, 2020; LaForce et al., 2019; Legewie & DiPrete, 2014).

2. CodeRVA as an inclusive STEM high school

CodeRVA's mission and program align with general descriptions of inclusive STEM schools in the literature, and in addition, its features are consistent with core characteristics identified for "exemplar" inclusive STEM high schools (LaForce et al., 2016; Lynch et al., 2018). For example, all students take advanced courses in computer science and related fields that prepare them for college and employment in STEM. The school maintains partnerships with businesses and academic institutions that support development of skills for the workplace, and it provides authentic STEM experiences through internships and other activities. Like most of the exemplar inclusive STEM schools (Lynch et al., 2018), CodeRVA is small, currently nearly at full size with 332 students, and diverse in terms of both socioeconomic status and race and ethnicity. The school's graduation rate for its first two cohorts was 100%, and although CodeRVA does not present its program as exclusively college preparatory, nearly two thirds of graduates planned to

attend a two- or a four-year college. On each of these indicators, CodeRVA falls well within the ranges reported for exemplar schools (Lynch et al., 2018).

2.1. Addressing barriers to access

During the years on which we focus, CodeRVA admitted students from fourteen participating school districts covering a geographic area of approximately 3500 square miles. As a magnet school designed to attract students across district boundaries, CodeRVA concentrates on removing barriers that might prevent students from accessing the opportunities it offers – including those that could discourage students from applying in the first place. To that end, the school provides bus transportation, laptop computers, and internet hotspots as needed. As is true for inclusive STEM schools generally (and in contrast to other regional magnet programs in Central Virginia), there are no entrance exams, math prerequisites, or other admissions criteria: application to CodeRVA requires only student (or parent) interest in the school.³ To apply, rising 9th graders complete a brief online application that enters them into an independently-operated lottery.⁴ Admission then takes place via weighted random selection, using weights for socioeconomic status (operationalized through a combination of family income, household size, and parent/guardian educational attainment) and gender to ensure that the student body reflects the composition of the region's population. Similar lottery approaches have been used elsewhere to achieve racial and economic diversity goals (e.g., Bhargava et al., 2008; Potter, et al., 2016).

We evaluated the success of the school's lottery admissions system in representing the region by comparing the composition of CodeRVA entering classes to the composition of 9th graders across the fourteen participating school districts. During 2018–2019 (the first year in which a full-size cohort of rising 9th graders was admitted), 41% of 9th grade students at CodeRVA were considered economically disadvantaged, close to the figure of 38% for participating districts as a whole. With no weights used to adjust directly for race or ethnicity, the CodeRVA entering cohort's composition approximated that of participating school districts, although non-Latinx Black and white students were somewhat overrepresented at CodeRVA (43% and 49% at CodeRVA vs. 34% and 46% in districts, respectively), while students who identified as Latinx, Asian, Pacific Islander, Native American, Alaskan Native, or multi-racial were underrepresented (8% at CodeRVA vs. 20% in districts). These distributions are encouraging in terms of using weights for socioeconomic status to increase access for minoritized students, though they also suggest that the school needs to enhance outreach to Latinx students and non-Black students of color (Corning et al., 2021).⁵

2.2. Efforts to provide a supportive environment

A second set of efforts by the school is geared toward creating an environment that helps students develop interest and engagement in computer science, supports strong academic outcomes, and encourages persistence in computer science or other STEM fields. These efforts fall into three areas, described below. Within each area, the school's efforts are consistent with critical components outlined in the literature on inclusive STEM schools (LaForce et al., 2016; Lynch et al., 2018).

2.2.1. High-quality materials and instruction

Opportunity gaps in STEM are associated with differences in socioeconomic status (Morgan et al., 2016, 2022), which may in turn be related to the quality of science facilities, the instructional approaches adopted, and/or accountability pressures (Bae & Lai, 2020; Hayes & Trexler, 2016; Oakes et al., 1990; Quinn & Cooc, 2015). Limited opportunity for involvement in authentic activities has the potential to hamper science engagement (Bae et al., 2021), while inquiry-based approaches support greater understanding of science content (Minner et al., 2010) and engagement with it (Bae et al., 2021). Experiential and

collaborative learning opportunities have also been linked to engagement in math and science (Fredricks et al., 2016; Fredricks et al., 2018), including by racially and ethnically minoritized students attending inclusive STEM schools (Bicer et al., 2020). Evidence points to classroom hands-on and social activities as mechanisms that explain the link between opportunities to participate in science and school engagement among students attending urban schools (Bae & Lai, 2020). In the area of computer science specifically, a National Academies of Science, Engineering, and Medicine report (NASEM, 2021b) recommended hands-on, authentic activities as an important avenue for enhancing students' interest in computing and feelings of belonging in the field.

CodeRVA's curriculum and program should promote student engagement. The school brings to bear sophisticated technology and incorporates elements of inquiry-based approaches and experiential learning. Project work, hands-on experiences, and internships are central features of the curriculum, and additional opportunities are provided through school special events. Support from a federal grant during the school's first several years facilitated acquisition of technology and provided for staff professional development related to curriculum design and the instructional approaches adopted.

2.2.2. Community characterized by shared interest and positive relationships

The computer science theme provides a focus for diverse guest speakers, field trips to businesses and colleges, and occasional computing-related special events, all of which should encourage a school "microclimate" (Aschbacher et al., 2010) or "motivational climate" (Robinson, 2023) that fosters students' computer science engagement and identity. When such school events and experiences lead to exploration and reflection by students, they may be especially effective in promoting STEM identity and motivation to persist (Perez et al., 2014). At the college level, learning communities designed to engage and support STEM students are associated with increased sense of belonging, academic integration, and intent to continue in the major, especially among historically underrepresented students (Xu et al., 2018; Solanki et al., 2019). At CodeRVA, student projects and especially the internships with industry partners, in which all students participate, create opportunities for collaborative work that explicitly makes connections to future careers. These activities may provide some of the social and cultural capital relevant to STEM careers that Bottia et al. (2021) find is important for supporting racially minoritized students' persistence in STEM.

Student surveys carried out as part of the school's evaluation process revealed strong student-teacher trust and high expectations for students (Corning et al., 2021) – elements that should help to satisfy students' needs for relatedness (Fredricks et al., 2011; Ryan & Deci, 2017) and developmental support, which are tied to engagement (Fredricks et al., 2019). In a climate of positive student-teacher relationships, inquiry-oriented approaches support hybrid spaces where academic curriculum can be integrated with funds of knowledge from students' homes and cultures (e.g., Bae, Sealy, Cabrera, Gladstone, & Mills, 2022; Calabrese Barton & Tan, 2009; Tan et al., 2023), creating avenues for students to bridge out-of-school identities with STEM identities. These kinds of opportunity structures grounded in interpersonal relationships and instructional activities that "uphold and reinforce students' esteemed cultural meaning systems" (Gray et al., 2018, p. 98) are important for fostering belonging among racially or ethnically minoritized students as well as female students, who may be likely to feel invisible, disconnected, or uncertain of whether they are valued in a particular setting (Rodriguez et al., 2022; Walton & Brady, 2017).

2.2.3. Personalized and blended learning

Finally, the school adopts a student-centered orientation that incorporates both personalized and blended learning – that is, learning that occurs through a combination of online and face-to-face instruction (see, e.g., Graham, 2013; Hrastinski, 2019; Rasheed et al., 2020).

Together, these approaches provide time and flexibility for teachers to work with small groups or individual students, and allow the school to offer a range of advanced options, including opportunities to earn college credits and an associate degree. Courses are sequenced so that none of the pathways depends on students entering high school with advanced math. The strength of a high school's STEM curriculum has been found to reduce the gender gap in plans to major in STEM in college, suggesting that high school opportunities to gain concrete experience with advanced work can be especially influential for disadvantaged and/or underrepresented groups (Legewie & DiPrete, 2014). The school's approaches also support students' needs for autonomy or choice in their learning, allowing them to interact with computer science in ways that they regard as personally meaningful and socially relevant (e.g., Ryoo et al., 2013; Kyte & Riegle-Crumb, 2017; Chang et al., 2014), and in turn, promoting engagement and motivation (Fredricks, 2011; Fredricks et al., 2019; Zumbunn et al., 2020).

Although we are not able to directly measure implementation of most of the elements of school context outlined above or link them directly to outcomes for CodeRVA and comparison group students, differences between CodeRVA's approach and that of traditional schools give us reason to investigate treatment-comparison group differences in three types of student outcomes, discussed below.

3. Potential impact of CodeRVA on course-taking, attendance, and test scores

3.1. Computer science course-taking

Recent national data on students' computer science experiences document legacies of historically unequal access to computer science education by race, gender, and socioeconomic status, and the same is true in Virginia. Just 66% and 73% of Virginia public high schools offered "foundational" computer science courses (courses that teach computer science principles and include at least 20 h of programming work per semester) in 2017–18 and 2018–19, respectively (Code.org et al., 2021). High schools with large percentages of economically disadvantaged students, as well as those in urban, town, or rural (in comparison to suburban) communities, were less likely to offer computer science courses (Code.org et al., 2021). Further influences contributing to persistent disparities include disciplinary norms and practices, cultural beliefs and stereotypes, educators' own biases about who belongs in computer science, scarcity of role models, and a lack of culturally relevant and responsive computer science instruction (Cheryan et al., 2012; Master et al., 2016; NASEM, 2021b). Indeed, even among Virginia high schools that offered computer science options, differential rates of AP test-taking reveal lopsided participation by gender and race: male students, and white and Asian students, were the most likely to take the exam (Code.org et al., 2020). NAEP data from 2017 show national race and locality disparities in access to "programming classes" among fourth and eighth graders (Von Zastrow, 2018), suggesting that these patterns may be in place early on.

The goal of inclusive STEM high schools is to increase student engagement and persistence in STEM, so one indication of success would be high levels of advanced course-taking on the part of attending students. Several studies have used administrative data or survey self-reports to examine the impact of inclusive STEM high schools on course-taking, finding positive effects on STEM and/or advanced math course-taking (Means et al., 2017; Saw, 2019). Students in our study were still in their first three years of high school, making it difficult to evaluate advanced course-taking, but we examine overall rates of computer science course-taking.

Computer science courses are required for all students attending CodeRVA, so high levels of computer science course-taking by admitted students are not in themselves an indication of student interest or achievement. Our focus here centers on levels of course-taking among the comparison group of applicants who were *not* admitted to CodeRVA

and instead attended traditional comprehensive schools. Since applicants were interested enough in computer science to apply to a computer-science-themed school, even those not selected for admission might go on to take computer science courses at their home district schools, reducing or even eliminating the effect of attending a specialized school. On the other hand, some research indicates that course-taking is not simply a function of course availability (Garland & Rapa-port, 2018; Thomas et al., 2013). A substantial difference between the two groups in levels of computer science course-taking would suggest that attendance at a specialized school shapes the content of students' educational experience quite directly.

In light of uneven availability of computer science course offerings in Virginia as a whole, we reviewed computer science course availability during the two years studied for students attending traditional comprehensive high schools in CodeRVA's participating region. In fact, all 40 of these traditional high schools in participating districts offered at least one foundational computer science course during 2017–2019 (data provided by the Virginia Department of Education [VDOE]).⁶ For 32 of those schools, we were able to obtain the districts' recent Program of Studies to verify that at least one foundational computer science course could be taken with no or minimal prerequisites (e.g., a keyboarding course or approval from a teacher) and/or during 9th or 10th grade.

3.2. School attendance

Inclusive STEM high schools are designed to boost learning in part by offering students a rich and appealing experience in terms of academic, co-curricular, and social life. Our school systems data do not include multidimensional measures of student engagement, but attendance data – which reflect the behavioral component of engagement (Fredricks et al., 2011) – are available for each student. Attendance can be seen as a minimum condition for engagement: students who are not present in school cannot be engaged. Moreover, missing school is associated with decreased school connectedness (Balfanz & Byrnes, 2012; Childs & Lofton, 2021; McNeely et al., 2002), and students' positive assessments of high school climate have been shown to be related to better attendance and reduced chronic absence, though the associations were small (Hamlin, 2021).

Attendance is also an important measure in its own right; increasing attention has focused on the negative effects of chronic absence in particular, and as of the 2018–2019 school year, Virginia's state accreditation standards include chronic absence as an indicator of school quality (VDOE, n.d.[a]). More frequent and/or chronic absence is associated with poorer academic performance (Gershenson et al., 2019; Gottfried, 2014), grade retention (Neild & Balfanz, 2006), greater risk of dropout (Balfanz et al., 2007; Rumberger & Thomas, 2000), and other negative long-term outcomes (Ansari & Pianta, 2019; Hallfors et al., 2002). Chronic absence has also been found to be higher among economically disadvantaged and racially or ethnically minoritized students (Balfanz & Byrnes, 2012; Gee, 2018; U.S. Department of Education, 2019), who may feel less connected to school (Gottfried et al., 2022), likely in part because the deficit models and white middle-class norms that often prevail can make schools inhospitable places (Grooms & Bohorquez, 2022). While much of the research has investigated the association between early-grade absence and high school outcomes, studies also find that decreases in attendance and increases in chronic absence are especially pronounced in high school (Balfanz & Byrnes, 2012; Miller & Johnson, 2016; Whitney & Liu, 2017), and that poor attendance in high school is associated with poor academic performance (Kirksey, 2019; Miller & Johnson, 2016) and lower likelihood of graduation (Allensworth et al., 2014; Kirksey, 2019; Lovelace et al., 2017–2018).

Within Virginia, Miller and Johnson (2016) identify three school districts as particularly high in chronic absence. Two of the three, Richmond City and Petersburg City, are participating districts for CodeRVA. These districts, together with a third, Hopewell City, had the

highest rates of poverty among CodeRVA's participating districts (23–24% for the period 2015–2019; U.S. Census Bureau, 2021), and very large percentages of economically disadvantaged students (54–75%), according to VDOE (n.d.[b]) for the school year covered by our attendance data, 2018–2019.⁷

As these data suggest, school engagement is just one of many factors – such as student well-being, living in poverty, and family and neighborhood challenges – that may influence school attendance and chronic absence (Balfanz & Byrnes, 2012; Childs & Lofton, 2021). Some research distinguishes between excused and unexcused (or voluntary) absences (Birioukov, 2016), with unexcused absences a measure that may better reflect academic or social disengagement as a response to the school environment – particularly for high school students who can decide whether or not to attend class (Gottfried, 2009; Liu & Loeb, 2021).

Characteristics of inclusive STEM high schools such as their small size, diverse student bodies, positive climate, and well-qualified teaching staff are consistent with best practices and with what is known about influences that bolster attendance (Chang & Romero, 2008; Liu & Loeb, 2021; Palardy et al., 2015; Tran & Gershenson, 2021), but effects on attendance have not been consistently found (Means et al., 2016). These features are all present at CodeRVA, but in addition, two further factors may support a positive attendance culture: the school enjoys close and trusting relationships between faculty and students (Corning et al., 2021), and school procedures include careful monitoring of attendance and quick intervention when problems are detected.

3.3. Standardized test scores

Finally, inclusive STEM high schools seek to provide an environment that supports robust academic outcomes and prepares all students to pursue STEM- or technology-oriented fields through postsecondary education and/or employment. Despite these goals, it is not clear how strongly inclusive STEM high schools are oriented toward enhancing STEM-specific academic performance. In their study of critical components of the models used by a sample of such schools, LaForce et al. (2016) found that school leaders tended to articulate their mission as centered on student engagement, development of critical thinking, and real-world problem-solving – rather than in terms of STEM content or disciplinary identity (LaForce et al., 2016). In fact, studies of inclusive STEM high schools' impact on math or science test scores have shown only limited evidence of positive effects (Means et al., 2021; Saw, 2019). Gnagey and Lavertu (2016) found positive effects for STEM subjects at just two of six schools studied; moreover, those positive effects appeared to come at the cost of non-STEM subject performance. Still, the elements of CodeRVA's approach to education outlined above and theoretical links to student engagement, motivation, and development of a computer science identity provide reason to investigate the school's potential effect on measures of academic learning. Thus – while bearing in mind that the school's primary focus is on supporting long-term computer science interest and persistence through a supportive environment – we examine CodeRVA's effect on students' academic performance, as measured through scores on standardized tests taken early in high school.

3.4. Effect on opportunity gaps

Because inclusive schools are intentionally designed to promote STEM interest and engagement among students who have historically had the fewest opportunities to participate, treatment effects might be greatest for those students. With respect to each of the above outcomes (computer science course-taking, attendance, and standardized test performance), we investigate whether CodeRVA helped to reduce gaps for students typically marginalized within STEM, examining differential effects by race, family income, and gender. We also investigate differential impacts for residence in urban and non-urban school districts; within CodeRVA's participating region, urban school districts include

those with both the highest population density and the highest poverty rates.

4. Data and method

We take advantage of CodeRVA's weighted random lottery admissions system, described above, which provides the framework for a quasi-experimental comparison of outcomes for students admitted to CodeRVA and those not admitted. For the cohorts entering during the two years examined, the number of applicants was four or more times the number of available seats. Data come from a restricted-use dataset obtained from the Virginia Longitudinal Data System (VLDS), with approval from the Virginia Department of Education (VDOE). This dataset allowed us to match students who applied for admission to CodeRVA with their stored school systems data; we draw on the lottery application data for supplementary information on family income. We use data on all applicants who applied to the school for entry during its first two years – 2017–18 and 2018–19 – and who could be successfully matched to school systems data (95.8%).⁸ Data were pooled for the two cohorts, for an analytic N of 985 students.

4.1. Outcome measures

For our investigation of student computer science course-taking, we used School Courses for the Exchange of Data (SCED) codes to match all computer science courses taken by applicants to a list of VDOE-identified foundational computer science courses (VDOE, n.d.[c]). Our course-taking outcome variable recorded whether or not students had taken at least one foundational computer science course during the 2017–2019 academic years.

To examine attendance, we focused on the September–March portion of the 2018–2019 academic year: approximately 106 school days for the majority of students in our treatment and comparison groups. We used three measures of attendance, each of which the literature suggests should reflect somewhat different aspects of engagement in school. The measure we label “attendance” represents days present as a percentage of total days of school (days present divided by the sum of days present plus days absent). We also examine “chronic absence,” adopting the definition used by Virginia and many other states to identify students absent during 10% or more of the total days of school. Finally, we examine “unexcused absence,” using unexcused days absent as a percentage of the total days of school, following a formula similar to the one above.

To evaluate the school's impact on academic outcomes, we used scores on standardized achievement tests administered each spring to assess the Virginia Standards of Learning (SOL). We consider results from seven of the eight main SOL tests taken by CodeRVA students in either their first or second year (2017–18 or 2018–19): Algebra I, Geometry, Biology, Chemistry, World History II, World Geography, and Writing.⁹ Given the limited number of CodeRVA students taking each test during the first two years, we pooled the test results across cohorts, but doing so does not meaningfully affect conclusions.

4.2. Analytic approach

Our approach to estimating treatment effects for CodeRVA included three steps. First, we assessed the balance of treatment and comparison groups on demographic and background variables, in order to determine the extent to which the lottery process facilitated the creation of two equivalent groups for comparison. Two different versions of a “treatment” effect are possible. Receiving an offer of admission to CodeRVA is the most methodologically rigorous version, since receiving the offer does not require any student action that could impact the quality of the randomization. Although defining treatment as enrollment in the school would focus more directly on the subset of admitted students who chose to accept the offer, doing so could bias treatment effects by introducing

Table 1

Applicant Demographic and Academic Characteristics, by Admission Status, 2017–2019 School Years.

Variable	Admitted N = 230	Not admitted N = 755	p
Female	116 (50.4%)	193 (25.6%)	< 0.001
Latinx	29 (12.6%)	55 (7.3%)	0.02
Black	102 (44.3%)	295 (39.1%)	0.17
White	105 (45.7%)	371 (49.1%)	0.37
Asian	5 (2.2%)	37 (4.9%)	0.09
Multi-racial or other race	18 (7.8%)	52 (6.9%)	0.47
Family income level	2.8 (1.5)	3.4 (1.4)	< 0.001
Prior math score	438.2 (56.9)	442.9 (58.9)	0.34
Prior science score	442.2 (50.8)	451.9 (53.0)	0.02
Prior writing score	442.5 (68.8)	453.5 (66.3)	0.03

Note. For categorical variables, Ns and percentages are presented. For continuous variables, means are presented, with standard deviations in parentheses. Family income represents gross income in six categories from below \$25,000 to \$200,000 or higher, but is treated as continuous.

unmeasured confounding variables (e.g., whether students had alternatives such as other magnet or honors schools to choose from). Therefore, treatment in the form of *admission* is our primary focus throughout, though we occasionally note the effect of *enrollment* in CodeRVA as well.¹⁰

Second, for each outcome examined, we conducted a series of unadjusted comparisons for students who did and did not receive the offer of admission. In these simple comparisons, any demographic or other differences between the treatment and comparison groups are confounded with differences in outcomes, yielding biased estimates of the impact of CodeRVA admission. These estimates, however, are useful as a starting point for comparison with our preferred estimates, obtained in the next step.

Our third step sought to account for observed and unobserved confounding variables through propensity score matching analysis (PSM; Abadie & Imbens, 2006; Guo & Fraser, 2014), for which we followed WWC guidelines (WWC, 2020). When properly specified to account for potential confounding variables, PSM has been shown to replicate effects obtained from experimental designs with true random assignment in educational research contexts (Cook et al., 2008; Pohl et al., 2009; Unlu et al., 2021). Our PSM analysis uses logistic regression to estimate a propensity score for each student from a number of baseline academic and demographic predictors. Students were then matched with counterparts in the comparison group with similar propensity scores, helping to eliminate potential bias in the groups' construction. PSM analyses were run in Stata version 17 (StataCorp, 2021) using the "teffects psmatch" routine with three matches, or nearest neighbors, per treatment group student. Matching with replacement was not permitted.

To assess the robustness of our results to model specifications and analytic approach, we compared results from this procedure to two other quasi-experimental analytic approaches commonly used in the field – augmented inverse probability weighting (AIPW; Cattaneo et al., 2013; Tan, 2010) and coarsened exact matching (CEM; Iacus et al., 2012). Estimates of the average treatment effect largely did not differ in terms of magnitude or significance; we discuss discrepancies below in the results section.

4.3. Covariates

To account for potential confounding factors, our models include several covariates: gender, self-identified race (represented in four categories: Black, white, Asian, and multi-racial or other identification), identification as Latinx, and family income (gross annual income represented in six categories ranging from below \$25,000 to \$200,000 or above, but treated as continuous). In the models predicting high school standardized test scores, we include students' earlier (middle school) standardized test outcomes in order to take account of prior opportunity

Table 2

Baseline Equivalence of Treatment and Comparison Groups, Before and After Matching.

Variable	Standardized Differences (d)		Variance Ratio	
	Raw	Matched	Raw	Matched
Female	0.62	-0.01	1.32	0.99
Latinx	0.20	0.01	1.85	1.04
Black	0.09	-0.10	1.01	0.98
White	-0.08	0.13	0.97	1.02
Asian	-0.11	-0.08	0.54	0.62
Multi-racial or other race	0.04	-0.08	1.14	0.75
Family income level	-0.55	0.02	0.91	1.10
Prior math score	-0.11	0.04	1.06	1.00
Prior science score	-0.19	-0.08	0.86	0.95
Prior writing score	-0.12	0.13	1.07	0.99

Notes. SOL = Virginia Standards of Learning test. Matching procedure used was (3) nearest neighbor without replacement.

and instructional quality.¹¹ Models for Algebra I and Geometry test scores include prior math SOL test scores, along with the other covariates. For Biology and Chemistry models, we include the same covariates, but substitute prior science SOL score for the prior math SOL score; similarly for Writing, we substitute prior writing score. For World Geography and World History II, no relevant prior score was available, so we include only the demographic covariates. The formal propensity score model used was as follows:

$$\text{Logit}(Y) = \beta_0 + \beta_1 \text{Gender}_i + \beta_2 \text{RaceEth}_i + \beta_3 \text{FamInc}_i + \beta_4 \text{PriorSOLScore}_i + \varepsilon_i$$

4.4. Balance of treatment and comparison groups

We began by reviewing the extent to which the lottery process facilitated the creation of equivalent treatment and comparison groups. Table 1 shows distributions on demographic and other characteristics of the admitted and non-admitted groups. These distributions were largely as expected: the treatment group was disproportionately female and less wealthy than the comparison group, reflecting the effect of the gender and socioeconomic status weights used to ensure that the admitted group represented the region.

The treatment and comparison groups also differed in two ways we did not anticipate, however. Admitted students were more likely to be Latinx, and admitted students had received lower scores on 7th grade science and writing SOLs (the two groups did not differ on middle school math scores). We attribute those differences to the fact that admitted students were more likely to be from lower socioeconomic status backgrounds, and that socioeconomic status is correlated with both ethnicity and prior achievement. Differences between the admitted and non-admitted groups suggest that adjusting for bias using PSM is important for improving the accuracy of estimated treatment effects.

4.5. Baseline equivalence and use of PSM

Our use of PSM adhered to the guidelines set forth in the What Works Clearinghouse (WWC) Standards and Procedures Handbook (v. 4.1, What Works Clearinghouse, 2020). The primary review criteria from the WWC for PSM, similar to other quasi-experimental methods, require that the analytic intervention and comparison groups are shown to have baseline equivalence. As shown in Table 2, after employing the PSM procedure, all observed differences between the two groups, including demographic and socioeconomic differences as well as differences in prior performance in math and science, were significantly reduced, from a range of $d = (-0.55, 0.62)$ in the unmatched sample to $d = (-0.10, 0.04)$ in the matched sample. The matched differences are within WWC guidelines for establishing baseline equivalence.

Table 3

Estimated Average Treatment Effect of Admission to CodeRVA on Computer Science Course-Taking, 2017–2019 School Years (PSM Analysis).

	ATE	A-I SE	z	p	LL	UL	Admitted N	Not admitted N
Admission	.352	.032	11.10	0.000	0.289	0.414	230	755

Note. ATE = Average Treatment Effect. A-I SE = Abadie-Imbens Robust Standard Error. LL and UL are the lower and upper limits of the 95% confidence interval. ATEs are log odds coefficients. P-values are based on a two-sample (binomial) test of proportions.

Table 4

Estimated Average Treatment Effect of Admission to CodeRVA on Three Measures of School Attendance, September – March, 2018–2019 School Year (PSM Analysis).

	ATE	A-I SE	z	p	LL	UL	Admitted N	Not admitted N
Attendance	1.11	.374	2.98	0.003	0.38	1.84	227	716
Unexcused absence	-.769	.195	-3.93	0.000	-1.15	-.385	212	707
Chronic absence	-.014	.017	-0.83	0.409	-.047	.019	230	755

Note. ATE = Average Treatment Effect. A-I SE = Abadie-Imbens Robust Standard Error. LL and UL are the lower and upper limits of the 95% confidence interval. ATEs for Attendance and Unexcused absence are shown as percentage points; ATEs for Chronic absence are log odds coefficients.

5. Results

5.1. Computer science course-taking

Our simple comparisons of computer science course-taking show that, among admitted students, 80% took a computer science course, compared to 48% of those not admitted ($p < .001$).¹² As anticipated, 100% of students who actually enrolled in CodeRVA took a computer science course. Since these comparisons do not take account of the systematic differences between the admitted and non-admitted groups, we next used the propensity matching procedure to obtain estimates of the average treatment effect (ATE) of admission to CodeRVA, controlling for confounding variables. Table 3 presents these results, including estimates, robust standard errors, and 95% confidence intervals.

Table 3 shows that the treatment effect on computer science course-taking is positive and statistically significant ($p < .001$) with potential confounding variables controlled. These results indicate that admission to CodeRVA increased the odds of taking a computer science course by 41%. (Enrollment in the school increased the odds by 69%).

We examined potential differential effects on course-taking by gender, race, family income, and residence in an urban or non-urban school district. Given the small cell sizes available for the investigation of differential effects, the gaps analyses here and below are necessarily exploratory. With respect to computer science course-taking, treatment effects were in the direction of reducing gaps; results for students in the second-lowest income group ($p = .04$) and for Black students ($p = .07$) were the most promising.¹³

5.2. School attendance

Our unadjusted simple mean comparisons showed no significant effect of admission on any of the attendance outcomes. After adjusting for differences between the treatment and comparison groups using PSM, however, we found that admission had a significant, positive effect on attendance and also significantly reduced unexcused absence (Table 4). For example, admission to CodeRVA was predicted to increase attendance by 1.11 percentage points, and to decrease unexcused absence by 0.769 percentage points. There was no effect of admission on chronic absence. (Enrollment had a significant positive effect on all three attendance outcomes after adjustment).

Since the maximum number of school days students could attend during the period examined was just over 100, the increase of about one percentage point shown for the attendance measure is equivalent to an increase of approximately one day. We calculated an approximate standardized effect size of 0.18 by dividing the ATE by the observed standard deviation for attendance. (With treatment defined as enrollment, the corresponding increase was about two percentage points,

Table 5

Mean Standards of Learning (SOL) Test Scores, by Admission Status, 2017–2019 School Years.

Variable	Admitted N = 230	Not admitted N = 755	p
Algebra I	436.9 (37.1)	434.7 (42.3)	0.59
Geometry	448.7 (47.7)	439.2 (47.3)	0.15
Biology	437.4 (47.9)	456.9 (51.7)	< 0.001
Chemistry	422.9 (45.1)	458.8 (44.0)	< 0.001
World Geography	441.0 (38.6)	461.2 (49.3)	< 0.001
World History II	442.1 (51.6)	450.8 (57.0)	0.22
Writing	467.2 (60.8)	462.9 (56.4)	0.61

Note. Means are presented, with standard deviations in parentheses.

equivalent to an increase of roughly two days, with an approximate effect size of 0.32).

We investigated whether admission to CodeRVA helped to reduce attendance gaps by students' race, family income, and urban/non-urban school district. The results showed no clear evidence of any differential impact by race or family income. However, when we considered students residing in the three urban school districts together, we found that the effect of admission to CodeRVA was significantly stronger for students from urban districts than for those from suburban/rural districts, reducing gaps in days attended ($p < .001$) and in unexcused absence ($p < .01$), with an effect on chronic absence in the same direction, though not significant ($p = .06$).

5.3. Standardized test (SOL) scores

Our unadjusted comparisons of high school standardized test scores for students who did and did not receive an offer of admission are presented in Table 5. Students admitted to CodeRVA had higher scores on all three math as well as the Writing standardized tests, compared to non-admitted students, although those differences did not reach statistical significance. On the other hand, admitted students had lower scores on the Biology, Chemistry, World Geography, and World History II tests.

Table 6 shows the results after using PSM to adjust for differences between the treatment and comparison groups. For example, the row for Algebra I shows that admission to CodeRVA is predicted to increase students' Algebra I score by 4.73 points, once other influences associated with admission are taken into account. The effects for Algebra I and Writing are positive in direction, but do not approach statistical significance. Effects for other test scores are negative in direction, with Biology and Chemistry showing differences significant at the $p < .05$ level. We again calculated approximate effect sizes by dividing the ATE by the observed standard deviation for each test showing a significant difference: these effect sizes were -0.25 (Biology) and -0.55

Table 6

Estimated Average Treatment Effects of Admission to CodeRVA on Standardized Test Scores, by SOL Test, 2017–2019 School Years (PSM Analysis).

Variable	ATE	A-I SE	z	p	LL	UL	Admitted N	Not admitted N
Algebra I	4.73	4.50	1.05	0.293	-4.89	13.57	142	278
Geometry	-4.71	5.93	0.79	0.427	-6.91	16.34	51	233
Biology	-12.80	4.28	-2.99	0.003	-21.19	-4.41	178	527
Chemistry	-34.36	9.72	-3.54	0.000	-53.40	-15.32	44	112
W. Geog.	-9.67	5.36	-1.80	0.071	-20.18	-0.84	119	102
W. Hist. II	-6.5	5.79	-1.12	0.261	-17.86	4.85	76	416
Writing	3.11	10.63	0.29	0.770	-17.732	23.946	50	365

Notes. ATE = Average Treatment Effect. A-I SE = Abadie-Imbens Robust Standard Error. LL and UL are the lower and upper limits of the 95% confidence interval.

(Chemistry).

Our exploration of potential differential impacts by race and socioeconomic status showed no statistically significant reduction in racial disparities in SOL test scores, and no clear picture of differential impact by socioeconomic status.

Several test outcomes (Algebra I, Geometry, and World History II) showed consistent gender differences in the impact of admission, with female students admitted to CodeRVA performing better than their peers. The interaction of gender by admission status was statistically significant only in one instance, which resulted in part from the fact that, although female students did somewhat better relative to their peers, male students admitted to CodeRVA performed more poorly than their counterparts at other schools.

Finally, we examined the potential differential impact of admission on students from urban districts as against those from suburban or rural districts combined. The direction of the effect on seven of the eight SOL outcomes was suggestive of greater impact for students from the urban districts, but none reached statistical significance.

5.4. Comparing PSM results with AIPW and CEM

To assess the robustness of our results to the choice of matching or weighting approach, we compared our PSM results to those obtained by two other common procedures, AIPW and CEM. A full comparison of estimates from all three approaches is presented in [Table A1](#), in the appendix. Overall, most estimates were similar in terms of magnitude, significance, and direction across all three approaches, and none of our major conclusions would be substantially altered if a different approach were used.

Still, we note several differences. The treatment effect estimate for Algebra I was positive and significant using CEM; this estimate was positive but not significant and of a similar magnitude and direction using PSM and AIPW. The treatment effect estimate for World Geography was negative and significant using AIPW; this estimate was negative but not significant and of a similar magnitude and direction using PSM and AIPW. The treatment effect estimate for chronic absence was negative and significant using PSM; this estimate was negative but not significant and somewhat different in magnitudes using PSM and AIPW. For attendance, the estimate was positive and similar in magnitude using all three methods, but was only significant for AIPW and PSM. Finally, for unexcused absences, the estimate was negative, similar in magnitude, and significant using AIPW and CEM, but was much smaller in magnitude and not significant using PSM.

6. Discussion

The results from this study contribute to research highlighting the potential of schools deliberately designed for equity and centered on STEM ([Means et al., 2017; 2021](#)). First, our finding that admission to an inclusive STEM high school has a positive impact on computer science course-taking serves as evidence that the school provides an educational experience substantially different from that of comparison group students, who attended traditional comprehensive schools. In terms of course availability alone, our research indicates that most of the

comparison students attending traditional schools had computer science courses available, yet fewer than half of them took those courses. Their lower levels of computer science course-taking may be a result of the choices they make or that are made for them, whether by school systems, personnel, or school culture. For instance, schools may not offer the courses every year for scheduling reasons or because of low enrollment. Some students may not be encouraged to take computer science and/or feel that they do not belong in a computer science class ([Cheryan et al., 2012; Eisenhart & Allen, 2020; Margolis et al., 2008; Master et al., 2016; NASEM, 2021b](#)). Additionally, although computer science courses count toward high school graduation requirements in Virginia, they do not satisfy higher education admission requirements ([Code.org & Alliance., 2021](#)). Further, since two of the most widely available computer science courses are offered as AP courses, they are vulnerable to staff assumptions (such as negative implicit biases) and school practices (such as tracking or gatekeeping policies) that, with regard to AP enrollment in general, have been found to disadvantage racially or ethnically minoritized students and those from lower-socioeconomic status backgrounds (e.g., [Lewis & Diamond, 2015; Margolis et al., 2008; Rodriguez & Hernandez-Hamed, 2020; Xu et al., 2021; Naff et al., 2021](#)).

We also considered the possibility that the limited nature of traditional schools' computer science course offerings, with no opportunity for further work or advancement, might reduce the appeal of those courses to students. However, among the high schools in CodeRVA's participating districts, 88% offered at least two courses, and 70% offered three or more – so most students attending traditional schools should have had the chance to undertake further work in computer science. These results underscore the potential value of inclusive, STEM-focused schools that offer a sequenced program with opportunities for advanced work, and cultivate a community of shared interest and awareness of the future opportunities STEM can open up ([Lynch et al., 2018; NASEM, 2021b](#)). Our findings support other evidence indicating that course-taking is not merely a matter of access ([Garland & Rapaport, 2018; Thomas et al., 2013](#)).

Our results showing the school's positive impact on attendance are also encouraging – particularly the significantly stronger effect for students from the three high-poverty, urban districts in our analysis. [Rumberger and Thomas \(2000\)](#) emphasize that characteristics of schools, not just characteristics of students, affect engagement and attendance, and best practices for promoting a school culture of attendance as well as more recent research findings underscore this point (e.g., [Chang & Romero, 2008; Gottfried & Hutt, 2019; Palardy et al., 2015; Tran & Gershenson, 2021](#)). Although our data do not allow us to connect CodeRVA's attendance gains to specific policies, we suspect that elements consistent with those best practices – supportive teacher-student relationships, an inclusive climate, a low student-teacher ratio, and responsive intervention and problem-solving by faculty and staff – contribute to a school environment where attendance problems are reduced.

On the other hand, we found no clear evidence of overall positive treatment effects on standardized test scores. It may be premature to expect effects on academic achievement, given that these data come from the first two years of CodeRVA's existence – a period of substantial change as the school expanded its faculty, increased its physical space

and facilities, developed its curriculum, and sought to reconcile an ambitious vision with operational realities. The rapidly evolving program and support system may have limited the school's impact.

Test score results may also have been affected by the tension – common to inclusive STEM high schools – between enhancing STEM curriculum and experiences and at the same time ensuring that students receive a traditional, broad academic grounding (Weis et al., 2015). Schools must make time in students' schedules for STEM-specific courses and electives, as well as internships and other activities designed to prepare students for STEM careers. Those challenges are compounded at a regional school where students depend on bus transportation, so that clubs and social activities must also be wedged into school hours. CodeRVA's strategy during the first two years was to condense the traditional three-year sequence of Biology, Chemistry, and Physics into two years – a practice the school has now decided to phase out. That integrated approach, coupled with an initial lack of appropriate science lab spaces and equipment that limited hands-on science experiences for students, may have affected test outcomes. Geography and History coursework has been similarly integrated with Language Arts, potentially affecting social science test scores.

A further possibility is that “distal” outcomes like state-level test scores may not be sensitive enough to detect treatment effects, especially compared to more proximal assessments such as open-ended performance-based tasks (Ruiz-Primo et al., 2002). Moreover, inclusive STEM high schools are often oriented toward providing students with a broader “life skills” base, rather than with math and science content knowledge (Gnagey & Lavertu, 2016; LaForce et al., 2016), and CodeRVA's website and promotional materials reveal a similar orientation. The school emphasizes the ways in which the program equips students with real-world skills and collaborative experiences that prepare them for college and career at least as much as it highlights its rigorous and advanced curriculum. Students' later outcomes, such as postsecondary enrollment and persistence in computer science, may be more aligned with the school's mission and represent better measures of its impact than performance on standardized tests.

6.1. Study limitations and directions for future research

This discussion of findings draws attention to several limitations of our study, as well as potential directions for future research. First, we recognize that our study examines the impact of just one inclusive school for computer science at a particular time and location; our results may not be generalizable to other settings, and our study does not have available measures of implementation of curriculum and program features. As part of a growing body of research, however, single-case studies like ours may expand opportunities for meta-analysis. The research team led by Means has studied multiple cohorts at inclusive STEM high schools in North Carolina, Texas, and Ohio, and their recent meta-analysis (Means et al., 2021) combines cohorts across those states, finding a consistently positive effect on several outcomes. They acknowledge, however, that all schools in their study received funding from the Bill & Melinda Gates Foundation, which may have contributed to similar designs, practices, and impact. A more comprehensive and robust meta-analysis might incorporate data from case studies like ours to reflect the effects of inclusive STEM schools established via separate processes in different states, and to explore variation in consistency of effects across school and state contexts.

Second, our positive results with respect to school attendance highlight the need for multi-dimensional measurement of engagement to help understand and evaluate the effects of inclusive STEM schools. Engagement's cognitive, emotional, and social dimensions are related to science learning opportunities and grades (Bae & Lai, 2020; Wang & Holcombe, 2010), and emotional engagement with science in particular shows a strong relationship to STEM career aspirations (Wang et al., 2016).

Third, although we regard our use of administrative data as a

strength of our study, it also restricts our investigation: we were able to assess just three types of outcomes, none of which fully reflects the computing-specific, experiential learning, or academic experiences the school seeks to provide. As the school matures and additional cohorts progress through CodeRVA and comparison schools, we will be able to extend the range of outcomes examined to include those expected in the second half of high school, such as graduation, college credits, and associate degrees earned. Much of the research investigating whether and why women and racially minoritized students remain in (or leave) STEM has focused on the college level (e.g., McGee, 2020; Perez et al., 2014; Xie et al., 2015), but there is a need to understand how middle- and high-school experiences contribute to or thwart STEM persistence. Our future analyses will provide an expanded portrait of the school's impact and should permit us to explore potential effects on opportunity gaps in greater detail.

6.2. Evaluation and policy implications

From an evaluation perspective, our study provides a careful analysis of students' experiences at an inclusive STEM school with respect to directly comparable students attending traditional comprehensive schools. The literature has defined characteristics of inclusive STEM high schools, but it is clear that there is wide variation in the degree to which individual schools exemplify the characteristics outlined (Lynch et al., 2018; Weis et al., 2015; Gnagey & Lavertu, 2016). When evaluations examine multiple schools at once, differences in quality of implementation may make it difficult to draw overall conclusions about the schools' effectiveness. Additionally, previous studies have assigned treatment at the school level, an approach that potentially confounds school and treatment effects and does not account for the possibility of self-selection bias when students (or their parents) choose particular schools. We add to this literature by employing a design where treatment assignment takes place at the level of students, guarding against self-selection bias. Our research illustrates the value of an approach that combines a quasi-experimental design for investigating school effectiveness with PSM to account for observed and unobserved group differences. Our adherence to WWC's PSM guidance should lend further strength to our conclusions.

From a policy perspective, these findings further affirm the potential of specialized STEM educational options – as long as they intentionally include mechanisms to increase equitable access. The long-standing lack of diversity along racial, gender, and socioeconomic lines at many of the nation's selective, STEM-specialized public high schools is an example of broader inequities in high-quality STEM learning experiences within the U.S. public education system (e.g., Morgan et al., 2016, 2022; Natanson, 2020, 2021; Gold et al., 2010). In spite of the current tech industry downturn, projected high demand for computer and information technology employees over the next decade (Bureau of Labor Statistics, 2022) will likely continue to fuel interest in STEM- and especially computer science-focused secondary education (Vegas et al., 2021). We do not suggest that our study school addressed the multifaceted ways in which the decision to apply to a particular school is encumbered by inequity (e.g., Phillipppo & Griffin, 2016). Nevertheless, the school's inclusive model bolstered access, equity, and diversity by drawing students across district boundaries; by eschewing merit-based selection processes and designing an admissions system to reflect regional socioeconomic and gender distributions; and by providing transportation and technology. Admitted students' attendance and exposure to foundational computer science courses were increased. We do not yet know how these students' careers will unfold, but since long-term persistence in computer science or STEM cannot occur without exposure and engagement, we believe these are essential first steps. Moreover, whether or not CodeRVA students persist in STEM, they will graduate with greater STEM/computer science literacy than students at traditional schools – consistent with one of the goals established for STEM education by the National Research Council (2011), and background

that will serve them well in the workplace and in life (Vegas et al., 2021).

While these are grounds for cautious optimism about the potential impact of inclusive STEM schools, larger questions remain about the quality of the education they provide, and the benefits that accrue to students. Some scholars have raised the question of whether “efforts to democratize STEM are plausible within our deeply stratified system” (Cipollone et al., 2020, p. 2). Equitable access is not sufficient if schools fail to equip all students to pursue high-level coursework, compromise on providing advanced content or core experiences due to financial, time, or accountability pressures, or do not support students in engaging with the dominant STEM or science discourse while sustaining their other identities (Allen & Eisenhart, 2017; Eisenhart & Allen, 2020; Weiss et al., 2015). When inclusive programs do not follow through on their promise, they risk providing students with “shadow capital” of limited value (Cipollone et al., 2020). Future research that attends to variation in inclusive STEM schools’ curriculum and instructional practices, considers implications for students’ racial/ethnic, gender, and intersectional identities, and evaluates graduates’ longer-term STEM outcomes is greatly needed.

Endnotes

¹ We use the term “opportunity gaps” rather than “achievement gaps” to highlight the fact that the gaps reflect differences in opportunities available to students throughout their educational careers, not merely differences in achievement at a particular point in time.

² In analyzing effects by urban/non-urban school district, our intent is to reflect the broad social and economic context, not to make assumptions about characteristics of the schools themselves, much less of their students. We recognize that conceptualizations of “urban education” often reflect a deficit perspective and an assumption that urban schools are monolithic (e.g., Milner, 2012; Welsh & Swain, 2020). For classifying participating school districts as “urban,” our main criterion was association with independent cities (rather than counties); among the districts included in our analysis, the city districts had the highest population density and the highest rates of poverty, and they share other systemic challenges related to education, health, etc.

³ Two participating school districts impose their own minimum eligibility criteria: one stipulates that applicants must have a minimum grade point average of 2.0, and the other deems ineligible applicants with a disciplinary infraction during the first semester of 8th grade.

⁴ For the first year (2017–2018) only, applications were accepted from both rising 9th and rising 10th graders, who were each admitted in half-size cohorts.

⁵ We use economic disadvantage (a standard measure used by the Virginia Department of Education to identify students eligible for free or reduced price lunch, or who show other indications of poverty, such as homelessness), in order to approximately evaluate the effect of the weighting. Lottery admissions were handled by an independent consultant using a proprietary algorithm, to which we do not have access. The importance of the weighted lottery adjustment is especially visible in terms of gender. Girls made up 49% of the regional cohort, but just 34% of applicants to the school; among CodeRVA 9th graders, weights restored the percentage of girls to 49%.

⁶ In 2017–18, one high school in a CodeRVA participating district did not offer any computer science courses, but it added a course the following year.

⁷ Among CodeRVA’s participating suburban/rural districts, only in Cumberland County did the poverty rate and percentage of economically disadvantaged students reach levels similar to those of the urban districts in our analysis. Thus the urban vs. non-urban distinction comes fairly close to, but does not completely coincide with, rates of poverty among CodeRVA’s participating districts. Poverty rates are 5-year estimates from the American Community Survey, 2015–2019 (U.S. Census Bureau, 2021), and economic disadvantage data are for the 2018–2019

Table A1

Comparison of Estimated Treatment Effects, by Analytic Approach.

	PSM		AIPW		CEM	
Variable	Estimate	<i>p</i>	Estimate	<i>p</i>	Estimate	<i>p</i>
CS course-taking	0.35	< .001	0.34	< .001	1.96	< .001
Algebra I	4.73	0.293	5.23	0.117	9.79	0.007
Geometry	-4.71	0.427	4.41	0.338	4.86	0.325
Biology	-12.80	0.003	-13.44	< .001	-7.41	0.012
Chemistry	-34.36	< .001	-29.19	< .001	-19.39	0.011
World Geography	-9.67	0.071	-10.73	0.040	-0.75	0.891
World History II	-6.50	0.261	-9.89	0.135	-5.18	0.503
Writing	3.11	0.770	6.62	0.398	19.06	0.052
Attendance	1.11	0.003	0.83	0.032	0.96	0.094
Chronic absence	-0.77	< .001	-0.01	0.687	-0.40	0.311
Unexcused absence	-0.01	0.409	-0.61	0.004	-0.71	0.015

Notes. PSM = nearest neighbor (3) propensity score matching without replacement. AIPW = Augmented inverse probability weighting. CEM = Coarsened exact matching.

school year (VDOE, n.d.[b]).

⁸ Those who could not be matched were primarily students who had been home-schooled or had attended private school, and as a result had no public school systems data record. If these applicants are excluded, the successful match rate increases to 98.09% for the remaining applicants. One participating school district did not successfully match any of its 10 applicants, so our analysis includes applicants from 13 of the 14 participating districts.

⁹ The only major SOL test taken by CodeRVA students during the first half of high school that we do not include in our analysis is Algebra II: for that test, the treatment group was very small, consisting of just 25 students. World History I is not offered at CodeRVA. Test scores exist for some students who retok the test while at CodeRVA, but because those scores are not reflective of CodeRVA’s curriculum or instruction, we do not examine them here.

¹⁰ Estimated effects when treatment is defined as enrollment are available from the authors upon request.

¹¹ We use standardized math and science test scores from 7th grade, since fewer students had valid 8th grade scores.

¹² Based on a two-sample (binomial) test of proportions.

¹³ Full results from analyses of differential effects by race, socioeconomic status, and gender are available from the authors upon request.

Author note

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Declaration of Competing Interest

None.

Data Availability

Detailed and raw data from the Virginia Longitudinal Data System are protected under the researcher’s contract and Restricted Use Data Agreement and are not available due to data privacy restrictions and

regulation, consistent with the Commonwealth of Virginia's Government Data Collections and Dissemination Practices Act.

Appendix

Table A1.

References

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